# Safety Data Sheets (SDSs)

For

Shenzhen Beizeer Electronics Co.,Ltd.

E District 3 Floor 1 Building Ink Factory Yingbo Dong Huan Two Road Longhua Street

Longhua New District Shenzhen Ctiy，China

and for their product

| | |
|---|---|
| Model /Type reference | 501035 |
| Trademark | BZ |
| Nominal Voltage | 3.7V |
| Typical Capacity | 155mAh 0.574Wh |
| Version Number | V1.0 |
| Preparation Date | Nov 11, 2019 |
| Revision Date | N/A |

## 1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

Product name: Li-ion Polymer Battery

**Other means of identification**

| | | |
|---|---|---|
| Product description | Model: | 501035 |
| | Trade mark: | BZ |
| | Nominal voltage: | 3.7V |
| | Typical capacity: | 155mAh 0.574Wh |
| | Weight: | 3.262g |

**Recommended use of the chemical and restrictions on use**

Recommended Use Used in portabl electronic equipments

Uses advidsed against:

a) Do not dismantle, open or shred secondary cells or batteries.
b) Do not expose cells or batteries to heat or fire. Avoid storage in direct sunlight.
c) Do not short-circuit a cell or a battery. Do not store cells or batteries haphazardly in a box or drawer where they may short-circuit each other or be short-circuited by other metal objects.
d) Do not remove a cell or battery from its original packaging until required for use.
e) Do not subject cells or batteries to mechanical shock.
f) In the event of a cell leaking, do not allow the liquid to come in contact with the skin or eyes. If contact has been made, wash the affected area with copious amounts of water and seek medical advice.
g) Do not use any charger other than that specifically provided for use with the equipment.
h) Observe the plus (+) and minus ( – ) marks on the cell, battery and equipment and ensure correct use.
i) Do not use any cell or battery which is not designed for use with the equipment.
j) Do not mix cells of different manufacture, capacity, size or type within a device.
k) Battery usage by children should be supervised.
l) Seek medical advice immediately if a cell or a battery has been swallowed.
m) Always purchase the battery recommended by the device manufacturer for the equipment.
n) Keep cells and batteries clean and dry.
o) Wipe the cell or battery terminals with a clean dry cloth if they become dirty.

p) Secondary cells and batteries need to be charged before use. Always use the correct charger and refer to the manufacturer’s instructions or equipment manual for proper charging instructions.
q) Do not leave a battery on prolonged charge when not in use.
r) After extended periods of storage, it may be necessary to charge and discharge the cells or batteries several times to obtain maximum performance.
s) Retain the original product literature for future reference.
t) Use only the cell or battery in the application for which it was intended.
u) When possible, remove the battery from the equipment when not in use.
v) Dispose of properly.

**Details of the supplier of the safety data sheet:**

| | |
|---|---|
| Supplier Name: | Shenzhen Beizeer Electronics Co.,Ltd. |
| Address: | E District 3 Floor 1 Building Ink Factory Yingbo Dong Huan Two Road Longhua Street Longhua New District Shenzhen Ctiy，China |
| Telephone number of the supplier: | 0755-83019227 |
| E-mail address: | 3424846643@qq.com |

**Emergency telephone number**

Company Emergency Phone Number: 18589062591

## 2. HAZARDS IDENTIFICATION

**Classification**

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200). This product is an article which is a sealed battery and as such does not require an MSDS per the OSHA hazard communication standard unless ruptured. The hazards indicated are

| | |
|---|---|
| Skin corrosion/irritation | Category 2 |
| Serious eye damage/eye irritation | Category 2 |
| Skin sensitization | Category 1 |
| Carcinogenicity | Category 2 |
| Specific target organ toxicity (repeated exposure) | Category 1 |

## GHS Label elements, including precautionary statements:

**Signal word** **Danger**

**Hazard Statements**
Causes skin irritation
Causes serious eye irritation
May cause an allergic skin reaction
Suspected of causing cancer
Causes damage to organs through prolonged or repeated exposures

This product is an article which contains a chemical substance. Safety information is given for exposure to the article as sold. Intended use of the product should not result in exposure to the chemical substance. This is a battery. In case of rupture: the above hazards exist.

**Appearance** Silver **Physical State** Solid containing liquid **Odor** None

**Precautionary Statements - Prevention**
Obtain special instructions before use
Do not handle until all safety precautions have been read and understood
Use personal protective equipment as required
Wash face, hands and any exposed skin thoroughly after handling
Contaminated work clothing should not be allowed out of the workplace
Wear protective gloves
Do not breathe dust/fume/gas/mist/vapors/spray
Do not eat, drink or smoke when using this product
Wear eye/face protection

**Precautionary Statements - Response**
IF exposed or concerned: Get medical advice/attention
Specific treatment (see supplemental first aid instructions on this label)

**Eyes**
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
If eye irritation persists: Get medical advice/attention

**Skin**
IF ON SKIN: Wash with plenty of soap and water
Take off contaminated clothing and wash before reuse
If skin irritation or rash occurs: Get medical advice/attention

**Precautionary Statements - Storage**
Store locked up

**Precautionary Statements - Disposal**
Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**
Not applicable
**Unknown Toxicity**
37.3% of the mixture consists of ingredient(s) of unknown toxicity

**Other information**
Very toxic to aquatic life with long lasting effects
Repeated or prolonged skin contact may cause allergic reactions with susceptible persons

**Interactions with Other Chemicals**
No information available.

## 3. COMPOSITION/INFORMATION ON INGREDIENTS

**Chemical characterixation: Mixtures**

**Description:**

Product: Consisting of the following components.

| Common Chemical Name | Concentration (%) | CAS Number |
|---|---|---|
| Cobalt lithium oxide | 39.8 | 12190-79-3 |
| Polyvinylidene fluoride | 0.71 | 24937-79-9 |
| Aluminum | 6.8 | 7429-90-5 |
| Graphite | 19.6 | 7782-42-5 |
| Styrene butadiene rubber | 0.52 | 61789-96-6 |
| Carbon Black | 0.83 | 1333-86-4 |
| Copper | 11.8 | 7440-50-8 |
| Nickel hydride | 0.39 | 14332-32-2 |
| Lithium hexafluorophosphate | 2.6 | 21324-40-3 |
| Ethylene carbonate | 4.2 | 96-49-1 |
| Ethyl methyl carbonate | 3.8 | 623-53-0 |
| Diethyl carbonate | 3.9 | 105-58-8 |
| Polyethylene | 3.3 | 9002-88-4 |
| Nylon | 0.35 | 25038-54-4 |
| polypropylene | 1.4 | 9003-07-0 |

Note: CAS number is Chemical Abstract Service Registry Number.

## 4. FIRST-AID MEASURES

**First aid measures**

First aid is upon rupture of sealed battery.

Eye Contact: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Seek immediate medical attention/advice.

Skin Contact: Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required. May cause an allergic skin reaction.

Inhalation: Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the

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|---|---|
| | aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. Get medical attention immediately if symptoms occur. |
| Ingestion | Do NOT induce vomiting. Rinse mouth immediately and drink plenty of water. Never give anything by mouth to an unconscious person. Call a physician or poison control center immediately. |
| Self-protection of the first cider | Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. Wear personal protective clothing (see section 8). |

**Most important symptoms and effects, both acute and delayed**

Itching. Coughing and/ or wheezing. Burning sensation.

**Indication of any immediate medical attention and special treatment needed**

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|---|---|
| Notes to Physician | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rates, frothy sputum, and high pulse pressure. May cause sensitization of susceptible persons.<br>Treat symptomatically. |

## 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**

$CO_2$, dry chemical powder, water spray.

Unsuitable Extinguishing Media:No information available.

**Specific Hazards Arising from the Chemical**

Under fire conditions, batteries may burst and release hazardous decomposition products when exposed to a fire situation.

Formation of toxic gases is possible during heating or in case of fire.

In case of fire, the following can be released:

Carbon monoxide(CO)

Carbon dioxide

Other irritating and toxic gases.

**Protective Equipment and Precautions for Firefighters**

Firefighters must wear fire resistant protective equipment and appropriate breathing apparatus. The staff must equip with filtermask (full mask) or isolated breathing apparatus. The staff must wear the clothes which can defense the fire and the toxic gas. Put out the fire in the upwind direction. Remove the container to the open space as soon as possible. Spray water on the containers in the fireplace to keep them cool until finish extinguishment.

# 6. ACCIDENTAL RELEASE MEASURES

**Personal precautions, protective equipment and emergency procedures**

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|---|---|
| Personal Precautions | Attention!Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| Other information | Refer to protective measures listed in Sections 7 and 8. |

**Environmental precautions**

Refer to protective measures listed in Sections 7 and 8. Prevent further leakage or spillage if safe to do so. Should not be released into the environment. Do not allow to enter into soil/subsoil. Prevent product from entering drains.

**Methods and material for containment and cleaning up**

| | |
|---|---|
| Methods for Containment | prevent further leakage or spillage if safe to do so. |
| Methods for Cleaning up | Use personal protective equipment. Dam up. Cover liquid spill with sand, earth or other Non combustible absorbent material. Pick up and transfer to properly labeled containers. Clean contaminated surface thoroughly. |

# 7. HANDLING AND STORAGE

**Precautions for safe handling**

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|---|---|
| Handling | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes and clothing. Wear personal protective equipment.<br>Wash thoroughly after handling. Use this material with adequate ventilation.<br>The product is not explosive. |

**Conditions for safe storage, including any incompatibilities**

If the Lithium-ion Battery is subject to storage for such a long term as more than 3 months, it is recommended to recharge the Lithium-ion Polymer Battery periodically.
3 months: -10℃~+40℃, 45 to 85%RH
And recommended at 0℃~+35℃ for long period storage.
The capacity recovery rate in the delivery state (50% capacity of fully charged) after storage is assumed to be 80% or more.
Do not storage Lithium-ion Battery haphazardly in a box or drawer where they may short-circuit each other or be short-circuited by other metal objects.
Keep out of reach of children.
Do not expose Lithium-ion Polymer Battery to heat or fire. Avoid storage in direct sunlight.
Do not store together with oxidizing and acidic materials.
Keep ignition sources away- Do not smoke.

Store in cool, dry and well-ventilated place.

**Incompatible Products** None known.

# 8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

### 9. Engineering Controls

Use local exhaust ventilation or other engineering controls to control sources of dust, mist, fumes and vapor. Keepaway from heat and open flame. Store in a cool, dry place.

**(b) Personal Protective Equipment**

**Respiratory Protection:** Not necessary under normal conditions.Skin and body Protection: Not necessary under normal conditions, Wear neoprene or nitrile rubber gloves ifhandling an open or leaking battery.

**Hand protection:** Wear neoprene or natural rubber material gloves if handling an open or leaking battery.

**Eye Protection:** Not necessary under normal conditions, wear safety glasses if handling an open or leaking battery.

**(c) Other Protective Equipment**

Have a safety shower and eye wash fountain readily available in the immediate work area.

**(d) Hygiene Measures**

Do not eat, drink, or smoke in work area. Maintain good housekeeping.

# 10. PHYSICAL AND CHEMICAL PROPERTIES

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|---|---|
| Physical State | Form: prismatic |
| | Color: Silver-white |
| | Odour: Odourless |
| | Odor Threshold: No information available |
| Change in condition: | |
| pH, with indication of the concentration | Not determined. |
| Melting point/freezing point | Not determined. |
| Initial boiling point and Boiling range: | Not determined. |
| Flash Point | Not determined. |
| Evaporation rate | Not determined. |
| Flammability (solid, gas) | Not determined. |
| Upper/lower flammability or explosive limits | Not determined. |
| Vapor Pressure: | Not determined. |
| Vapor Density: | Not determined. |
| relative density: | Not determined. |

| | |
|---|---|
| Solubility in Water: | Not determined. |
| Solubility in other solvents | Not determined. |
| n-octanol/water partition coefficient | Not determined. |
| Auto-ignition temperature | Product is not self-igniting. |
| Decomposition temperature | Not determined. |
| Odout threshold | Not determined. |
| Evaporation rate | Not determined. |
| Viscosity | Not determined. |
| Other Information | No further relevant information available. |

## 11. STABILITY AND REACTIVITY

**Reactivity:** Stable under recommended storage and handling conditions (see section 7, Handling and storage).
**Chemical stability:** Stable under normal conditions of use, storage and transport.
**Thermal decomposition/conditions to be avoided:** No decomposition if used according to specifications.
**Possibility of Hazardous Reactions:** None under normal processing.
**Hazardous Polymerization:** Hazardous polymerization does not occur.
**Conditions to avoid:** Strong heating, fire, Incompatible materials.
**Incompatible materials:** Strong oxidizing agents. Strong acids.Base metals.
**Hazardous Decomposition Products:**Carbon oxides, Other irritating and toxic gases.

## 12. TOXICOLOGICAL INFORMATION

**Acute toxiciy:** No data available.

| LD/LC50 values relevant for classification: |
|---|
| Not available. |

**Skin corrosion/irritation:** No irritant effect.
**Serious eye damage/irritation:** Cause serious eye irritation.
**Respiratory or skin sensitization:** No sensitizing effects known.
**Specific target organ system toxicity:** No information available.
**CMR effects(carcinogenity, mutagenicity and toxicity for reproduction):** No information available.

## 13. Ecological Information

**Toxicity:**

| Acquatic toxicity: |
|---|

No further relevant information available.

**Persistence and degradability:** No further relevant information available.

**Bioaccumulative potential:** No further relevant information available.

**Mobility in soil:** No further relevant information available.

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

**Other adverse effects:** No information available.

## 14. DISPOSAL CONSIDERATIONS

**Waste treatment methods**

Recommendation: Must not be disposed together with household garbage.

Do not allow product to reach sewage system

**Uncleaned packaging:**

Recommendation: Disposal must be made according to official regulations.

## 15. TRANSPORT INFORMATION

**Land transport**

ADR/RID class: Not regulated.

UN-Number: UN3480 or UN3481.

**Maritime transport**

IMDG Class: Class 9.

UN Number: UN3480 or UN3481.

Marine pollutant: No

**Air transport**

ICAO/IATA Class: Class 9

UN/ID Number: UN3480 or UN3481

Environmental hazards: Not applicable.

Special precautions for user: Not applicable.

Transport/Additional information: Not restricted goods according to the above specifications.

The Lithium-ion Battery had been tested according to the requirements of the UN manual of tests and Criteria, Part III, subsection 38.3;

The lithium ion batteries according to Section II/Section IB of PACKING INSTRUCTION 965, or Section II of PACKING INSTRUCTION 966～967 of the Dangerous Goods regulations 58th Edition may be transported.

The packaging shall be adequate to avoid mechanical damage during transport, handling and stacking. The materials and pack design shall be chosen so as to prevent the development of unintentional electrical conduction, corrosion of the terminals and ingress of moisture.

Meets requirements of DOT Special Provision 188 to be transported as non-dangerous goods

Meets the requirements of 49CFR173.185 to be transported as non-dangerous goods for road, rail, air, and vessel (Effective August 6, 2014 per HM224F)
The package must be handled with care and that a flammability hazard exists if the package is damaged;

## 16. REGULATORY INFORMATION

**OSHA hazard communication standard (29 CFR 1910.1200)**

**_____Hazardous V _____Non-hazardous**

## 17. OTHER INFORMATION

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

| **Preparation and revision information** | | |
|---|---|---|
| Date of previous revision | Date of this revision | Revision summary |
| Not applicable | | The first New SDS |

**Relevant phrases:**

R20/22: Harmful by inhalation and if swallowed.
R36: Irritating to eyes.
H302: Harmful if swallowed.
H332: Harmful if inhaled.

**********End of SDS**********